MAGNATE ACADEMY

TECHNICAL ROUND 2 – DATA SCIENCE

Candidate Name:
Date: **Total Marks: 50**

Instructions

- Answer all questions.
- Write Python code wherever requested.
- Give clear explanations and examples where appropriate.
- The practical task will be evaluated separately as a complete Data Science workflow.

PART 1 – PYTHON | 15 MARKS

A. Python Coding | 10 Marks

1. Largest and Smallest Number — 2 Marks

Write a Python program to find the **largest and smallest numbers** in a given list.

2. First Non-Repeating Character — 2 Marks

`Write a Python program to find the **first non-repeating character** in a string.`

`Example:`

```
Input:  "aabbcdde"

Output: "c"
```

3. Anagram Check — 2 Marks

Write a Python program to check whether two strings are **anagrams** of each other.

Condition: Do not use the built-in `sorted()` function.

Example:

Input:

"listen"

"silent"

Output:

Anagram

4. Merge Two Dictionaries — 2 Marks

Write a Python program to merge two dictionaries.

If both dictionaries contain the same key, add their values.

Example:

dict1 = {"a": 10, "b": 20}

dict2 = {"b": 30, "c": 40}

Expected result:

{"a": 10, "b": 50, "c": 40}

5. Find the Missing Number — 2 Marks

Given a list containing `n` distinct numbers taken from the range `0` to `n`, write a Python program to find the missing number.

Example:

Input: [3, 0, 1]

Output: 2

B. Python Theory | 5 Marks

1 What are `*args` and `**kwargs` in Python? Give a suitable example
2 What is the difference between a **parameter** and an **argument** in Python?

3 What is the difference between `print` and `return` in Python?
4 What is the `range()` function in Python? Give an example.
5 What are **nested conditions** in Python? Give a suitable example.

PART 2 – MACHINE LEARNING | 5 MARKS

1 What is **Overfitting** in Machine Learning? How can it be reduced?
2 What is **Underfitting** in Machine Learning?
3 What is the difference between **Parameters** and **Hyperparameters** in Machine Learning?
4 Differentiate between **Hard Voting** and **Soft Voting** in Ensemble Learning.
5 What is **Boosting**? How is it different from **Bagging**?

PART 3 – MACHINE LEARNING PRACTICAL | 15 MARKS

Complete Data Science Project

Task

Download a **new and suitable dataset** and build a complete Machine Learning project.

Your project must include:

Requirement	Marks
Dataset selection and loading	2
Data preprocessing	3
Exploratory Data Analysis and visualisation	2
Feature/target selection and train-test split	2
Model selection and training	2
Model evaluation and interpretation	2
GitHub repository and project documentation	2
Total	**15**

GitHub Requirement

Create a GitHub repository named : **`Technical round 1`**

Push the complete project to the repository, including the **source code, preprocessing, visualisations, model training, evaluation and `README.md`**.

> **Note: All stages must be completed in the given order. If any stage is skipped, the marks for that stage and all subsequent stages will be zero, even if the subsequent stages are completed. For example, skipping Stage 3 results in 0 marks for Stages 3–7.**

PART 4 – DEEP LEARNING | 10 Marks

1 What is **Deep Learning**?
2 What is an **Activation Function**? Why is it used in a neural network?
3 What is **Forward Propagation**?
4 What is **Backward Propagation (Backpropagation)**?
5 What are the default values of **`epochs`** and **`batch_size`** when using Keras `model.fit()` if they are not explicitly specified?
6 What is a **Convolution Operation** in a Convolutional Neural Network (CNN)?
7 What is **Pooling** in CNN? Why is it used?
8 What is **Sequential Data**? Give one suitable example.
9 What is the difference between a **CNN** and an **RNN**?
10 What is a **Neuron** in a Neural Network?

PART 5 – NLP | 5 Marks

1 What is **Natural Language Processing (NLP)**?
2 What are some common **applications of NLP**?
3 When is **Sentence Tokenisation** useful?
4 What is **Tokenisation** in NLP?
5 When is **Word Tokenisation** useful?